

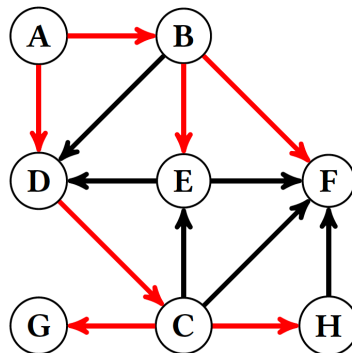
Advanced Algorithms and Data Structures

Graph

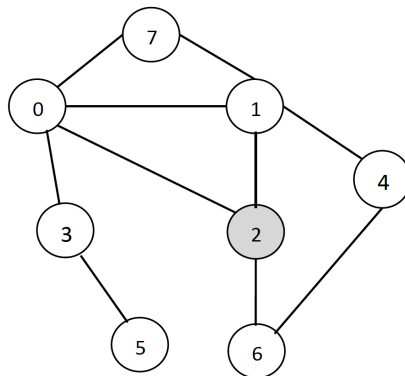
Exercises

Prepared by LY Rottana

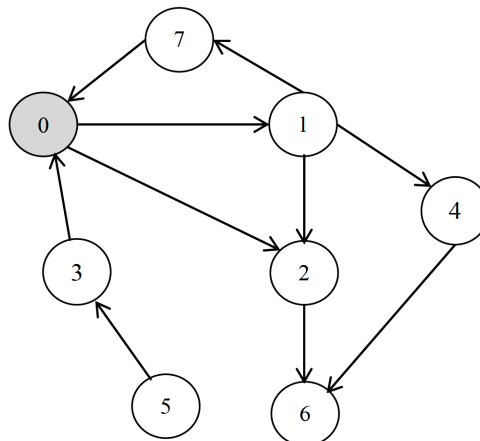
1. Execute a breadth-first search and a depth-first search on the following (directed) graph starting from vertex A. When processing the neighbors of a vertex, process them in alphabetical order.



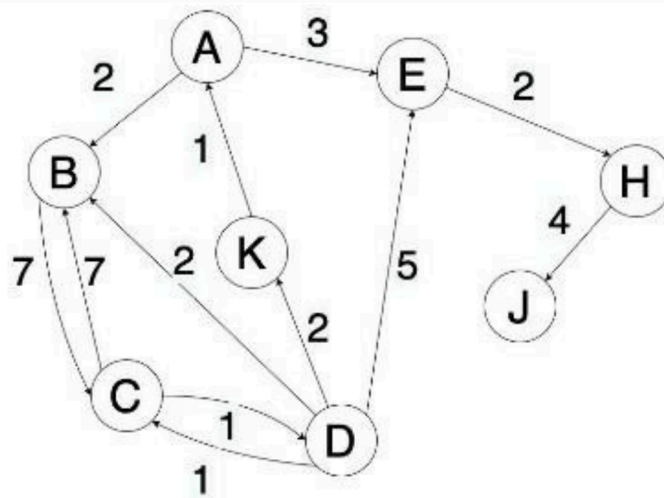
2. Traverse the graph below in BFS order, starting from vertex 2. Whenever you examine neighbors of a vertex, process them in increasing order of vertex number.



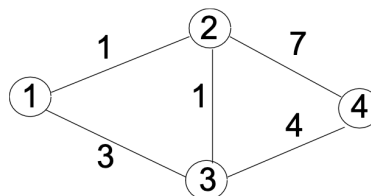
3. Traverse the graph below in DFS order, starting from vertex. Whenever you examine neighbors of a vertex, process them in increasing order of vertex number.



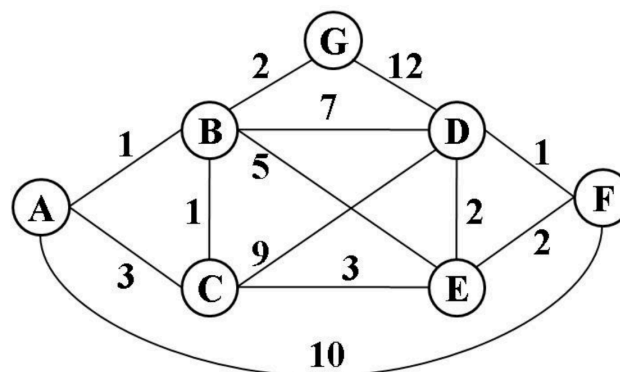
4. Given the following graph, simulate the breadth first graph search algorithm starting with vertex A. Write down one possible order of vertices that would be processed by the BFS traversal.



5. Given the same graph, simulate the depth first graph search algorithm starting with vertex A. Write down one possible order of vertices that would be processed by the DFS traversal.
6. Consider the following graph. Use Dijkstra's algorithm to find the shortest 1 – 4 path.



7. Consider the following undirected, weighted graph. Step through Dijkstra's algorithm to calculate the single-source shortest paths from A to every other vertex. After that, indicate the lowest-cost path from node A to node F.



8. Given the following map as the network of routes. The length from each point to another point is represented in km.
- Use Dijkstra's algorithm to find the shortest route from A to J. State your shortest route and its length.
 - The road from C to F will be closed next week for repairs. Find a shortest route from A to J that does not include CF and state its length.

